
MODULE *appex4_1_5*

EXTENDS *Naturals, Integers, TLC*

CONSTANTS $x0, y0, min, max$

ASSUME $x0 \in min .. max \wedge y0 \in min .. max$

--algorithm *commeonveut*{

variables $x = x0,$
 $y = y0,$
 $z;$

{

$l0:$ **assert** $x = x0 \wedge y = y0;$

if ($x < y$) {

$l1:$ **assert** $x = x0 \wedge y = y0 \wedge x < y;$

$z := y;$

$l2:$ **assert** $x = x0 \wedge y = y0 \wedge x \leq z \wedge y \leq z \wedge z \in \{x, y\};$

}

else

{

$l3:$ **assert** $x = x0 \wedge y = y0 \wedge x \geq y;$

$z := x;$

$l4:$ **assert** $x = x0 \wedge y = y0 \wedge z = y0 \wedge x \leq z \wedge y \leq z \wedge z \in \{x, y\};$

} ;

$l5:$ **assert** $x = x0 \wedge y = y0 \wedge z \in \{x, y\} \wedge x \leq z \wedge y \leq z;$

print "done";

$l6:$ **assert** $x = x0 \wedge y = y0 \wedge z \in \{x, y\} \wedge x \leq z \wedge y \leq z;$

}

}

BEGIN TRANSLATION

CONSTANT *defaultInitValue*

VARIABLES x, y, z, pc

$vars \triangleq \langle x, y, z, pc \rangle$

$Init \triangleq$ Global variables
 $\wedge x = x0$
 $\wedge y = y0$
 $\wedge z = defaultInitValue$
 $\wedge pc = "l0"$

$l0 \triangleq$ $\wedge pc = "l0"$
 $\wedge Assert(x = x0 \wedge y = y0, \text{"Failure of assertion at line 16, column 5."})$
 $\wedge \text{IF } x < y$
THEN $\wedge pc' = "l1"$

$$\begin{aligned}
& \text{ELSE } \wedge pc' = \text{"l3"} \\
& \wedge \text{UNCHANGED } \langle x, y, z \rangle \\
l1 & \triangleq \wedge pc = \text{"l1"} \\
& \wedge \text{Assert}(x = x0 \wedge y = y0 \wedge x < y, \\
& \quad \text{"Failure of assertion at line 18, column 8."}) \\
& \wedge z' = y \\
& \wedge pc' = \text{"l2"} \\
& \wedge \text{UNCHANGED } \langle x, y \rangle \\
l2 & \triangleq \wedge pc = \text{"l2"} \\
& \wedge \text{Assert}(x = x0 \wedge y = y0 \wedge x \leq z \wedge y \leq z \wedge z \in \{x, y\}, \\
& \quad \text{"Failure of assertion at line 20, column 7."}) \\
& \wedge pc' = \text{"l5"} \\
& \wedge \text{UNCHANGED } \langle x, y, z \rangle \\
l3 & \triangleq \wedge pc = \text{"l3"} \\
& \wedge \text{Assert}(x = x0 \wedge y = y0 \wedge x \geq y, \\
& \quad \text{"Failure of assertion at line 24, column 6."}) \\
& \wedge z' = x \\
& \wedge pc' = \text{"l4"} \\
& \wedge \text{UNCHANGED } \langle x, y \rangle \\
l4 & \triangleq \wedge pc = \text{"l4"} \\
& \wedge \text{Assert}(x = x0 \wedge y = y0 \wedge z = y0 \wedge x \leq z \wedge y \leq z \wedge z \in \{x, y\}, \\
& \quad \text{"Failure of assertion at line 26, column 6."}) \\
& \wedge pc' = \text{"l5"} \\
& \wedge \text{UNCHANGED } \langle x, y, z \rangle \\
l5 & \triangleq \wedge pc = \text{"l5"} \\
& \wedge \text{Assert}(x = x0 \wedge y = y0 \wedge z \in \{x, y\} \wedge x \leq z \wedge y \leq z, \\
& \quad \text{"Failure of assertion at line 28, column 6."}) \\
& \wedge \text{PrintT}(\text{"done"}) \\
& \wedge pc' = \text{"l6"} \\
& \wedge \text{UNCHANGED } \langle x, y, z \rangle \\
l6 & \triangleq \wedge pc = \text{"l6"} \\
& \wedge \text{Assert}(x = x0 \wedge y = y0 \wedge z \in \{x, y\} \wedge x \leq z \wedge y \leq z, \\
& \quad \text{"Failure of assertion at line 30, column 6."}) \\
& \wedge pc' = \text{"Done"} \\
& \wedge \text{UNCHANGED } \langle x, y, z \rangle
\end{aligned}$$

Allow infinite stuttering to prevent deadlock on termination.

$\text{Terminating} \triangleq pc = \text{"Done"} \wedge \text{UNCHANGED } vars$

$\text{Next} \triangleq l0 \vee l1 \vee l2 \vee l3 \vee l4 \vee l5 \vee l6$
 $\vee \text{Terminating}$

$$Spec \triangleq Init \wedge \Box[Next]_{vars}$$

$$Termination \triangleq \Diamond(pc = \text{"Done"})$$

END TRANSLATION

$$maximum(X, Y, Z) \triangleq Z \in \{X, Y\} \wedge X \leq Z \wedge Y \leq Z$$

$$\begin{aligned}
Inv \triangleq & \\
& \wedge pc \in \{\text{"l0"}, \text{"l1"}, \text{"l2"}, \text{"l3"}, \text{"l4"}, \text{"l5"}, \text{"l6"}, \text{"Done"}\} \\
& \wedge x \in Int \wedge y \in Int \wedge (z \neq defaultInitValue \Rightarrow z \in Int) \\
& \wedge pc = \text{"l0"} \Rightarrow x = x0 \wedge y = y0 \\
& \wedge pc = \text{"l1"} \Rightarrow x = x0 \wedge y = y0 \wedge x < y
\end{aligned}$$

$$safetyrte \triangleq x \in min .. max \wedge y \in min .. max \wedge z \in min..max$$

$$safetypcv1 \triangleq pc = \text{"Done"} \Rightarrow x = x0 \wedge y = y0 \wedge z \in \{x, y\} \wedge x \leq z \wedge y \leq z$$

$$safetypcv2 \triangleq pc = \text{"Done"} \Rightarrow x = x0 \wedge y = y0 \wedge maximum(x, y, z)$$

\ * Modification History
\ * Last modified *Tue Oct 04 09:20:27 CEST 2022* by *mery*
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